

Quick Recall (Ch. 7.2)

Set systems & antichains (Def). For a finite set X with $|X| = n$, a *set system* $\mathcal{M} \subseteq 2^X$ is called *independent* (or an *antichain*) if it contains no two distinct sets A, B with $A \subset B$.

Chains. A *chain* in 2^X is a family $\{A_1, \dots, A_k\}$ with $A_1 \subset A_2 \subset \dots \subset A_k$. A *maximal chain* adds elements one at a time:

$$\emptyset \subset \{x_1\} \subset \{x_1, x_2\} \subset \dots \subset X.$$

Maximal chains are in bijection with linear orderings of X – exactly $n!$ of them.

Key observation. Any chain meets any antichain in at most one set.

Sperner's Theorem (Thm 7.2.1). Every antichain $\mathcal{M}$ on an n -element set satisfies

$$|\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Equality is attained by $\mathcal{M} = \binom{X}{\lfloor n/2 \rfloor}$.

Proof sketch (chain counting). Count pairs (R, M) with $M \in \mathcal{M}$ and R a maximal chain through M :

- Each chain meets $\mathcal{M}$ at most once $\Rightarrow \#(R, M) \leq n!$.
- Each $M \in \mathcal{M}$ lies on $|M|!(n - |M|)!$ maximal chains.

Combining gives the LYM inequality

$$\sum_{M \in \mathcal{M}} \binom{n}{|M|}^{-1} \leq 1,$$

hence $|\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

Practice: Textbook Exercise 7.2.6* — the Littlewood–Offord problem, part (a)

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ with $|a_i| \geq 1$ for every i . For each sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, +1\}^n$, define

$$S(\varepsilon) := \sum_{i=1}^n \varepsilon_i a_i, \quad p(a_1, \dots, a_n) := \#\{\varepsilon \in \{\pm 1\}^n : -1 < S(\varepsilon) < 1\}.$$

(a)* Prove that for any choice of $a_1, \dots, a_n$ with $|a_i| \geq 1$,

$$p(a_1, \dots, a_n) \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Hints

Hint (textbook). Apply Sperner's theorem on the set system $\{\{i : \varepsilon_i = 1\} : \sum_i \varepsilon_i a_i \in (-1, 1)\}$.

Suggested steps. (1) WLOG assume $a_i > 0$ for all i (use a sign-flip bijection on $\{\pm 1\}^n$). (2) For each valid ε , set $S_\varepsilon := \{i : \varepsilon_i = +1\}$. (3) Show $\mathcal{M} := \{S_\varepsilon : \varepsilon \text{ valid}\}$ is an antichain by computing $S(\delta) - S(\varepsilon)$ when $S_\varepsilon \subsetneq S_\delta$.

 Practice: Textbook Exercise 7.2.6 — the Littlewood–Offord problem, part (b)

(Setup as in part (a): $a_1, \dots, a_n \in \mathbb{R}$ with $|a_i| \geq 1$, $S(\varepsilon) := \sum_i \varepsilon_i a_i$, and $p(a_1, \dots, a_n) := \#\{\varepsilon \in \{\pm 1\}^n : -1 < S(\varepsilon) < 1\}$.)

(b) Find $a_1, \dots, a_n$ with $p(a_1, \dots, a_n) = \binom{n}{\lfloor n/2 \rfloor}$.

 Hints

Hint. Try the simplest possible choice: $a_1 = a_2 = \dots = a_n = 1$. For which parity of n does this attain the bound, and why?